



ရန်ကုန်အရှေ့ပိုင်းတက္ကသိုလ်၊ ၂၀၂၆ ပညာသင်နှစ်၊ စာစဉ်အဖြေသွားစာအုပ်

အမည်	- မောင်အောင် (ဖို.ပိုင်)
ခုံအမှတ်	- ၃-၇၈-၁၆ (၅၆)
အထူးပြု	- ရုက္ခဗေဒ
Module No./Module Name	- Bot - 2104 / (Ecology)
Assignment I / II	- I

1. Four natural units of ecology

- (1) Individual
- (2) Population
- (3) Community
- (4) Ecosystem

Community

- Associated organisms having natural relationship to each other and to their environment are recognized as community.
- To study community all of the populations of organisms (plants, animals, and decomposers) that exists together in a given local area should be studied.
- At the community level, ecologists study interactions between populations; how they coexist and relate to one another in terms of obtaining items necessary for survival, such as food, shelter, and mate.
- Ecology Ecological communities are not static but change over time in response to climatic changes, human intervention, and geological events.

2. - Water is always presents in the atmosphere in the form of invisible water vapor.
- This is normally known as humidity of the air. The quality of water vapor present in the atmosphere at a particular place depends on several factors.
 - Water vapor in the atmosphere generally is measured as the relative humidity (RH), defined as the percentages of water in the air relative to the amount of water that the air could hold were it fully ~~saturated~~ saturated at a given temperature.
 - Warm air holds more moisture (has a higher RH) than cool air. RH varies according to differences in topography, vegetation, ~~and~~ elevation and microclimate.
 - Temperature and wind both affect evaporation from plant and soil.

3. - It is the relationship between the individuals of different species. Individuals of one species suffer a reduction in fecundity, survivorship, or growth as a result of exploitation of resources or interference by individuals from another species.
- The interrelationship may be either temporary or lifelong. It may be beneficial, harmful or without any apparent effect.
 - These competitive effects on individuals are likely to affect the population dynamics of the competing species.
 - These, in turn, can influence the species' distributions, and also their evolution.
 - The distributions and abundances of species, of course, determine the compositions of the biological communities of which they are part.



အမည်

- မောင်မောင်မြိုး

ခုံအမှတ်

- ၃-၇၁-၁၆ (၁၅၆)

အထူးပြု

- ရုက္ခဗေဒ

Module No./Module Name

- Bot-2104 / (Ecology)

Assignment I / II

- II

1. Effect on Photosynthesis

- Photosynthesis is the only efficient means by which energy from abiotic world is trapped for utilization by the biotic world.
- The rate of photosynthesis varies somewhat with different waves lengths of light.
- In terrestrial ecosystems the quality of light does not change enough to have an important differential effect on the rate of photosynthesis.
- But in aquatic ecosystem as light penetrates water, the red and blue colors of the light are filtered out and the resultant greenish light is poorly absorbed by chlorophyll.
- However, the red algae living at greater depths are able to utilize the light energy due to phycoerythrin.

Effect on Growth

- The growth of plants is also greatly influenced by the intensity and direction of light.
- Light is necessary for growth, yet stems show greater elongation in the darkness than in the light. More rapid growth takes place at night.
- In total absence of light for a long time plants become etiolated with weak, pale yellow stem in which tissues are poorly differentiated and mechanical tissues are lacking.
- The branches are few and leaves become scaly. Roots are poorly developed. Plants ~~grown~~ grown in strong light remain short and stocky and present a healthy look.